Relational Inductive Bias in an Event-Centric Crime Knowledge Graph

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1 Introduction

Knowledge graphs provide a natural way to represent relational domains, where entities are connected through typed relations rather than treated as independent rows in a table. Crime incident data is one example of such a domain - an incident can involve multiple victims, offenders, offenses, victim types, offense types, and victim-offender relationship records. Representing this data as a knowledge graph allows a model to perform message passing over these relations and potentially use this structure that is lost in a flat tabular representation.

To explore whether the graph structure of a KG is useful, a crime knowledge graph is used in this project. The graph contains many entities, such as victims, offenders, and incidents, that are unique to a single event or appear only in a small local neighborhood. This limits the amount of reusable entity-level structure (such as repeated crimes for a certain offender) available to a graph neural network. For such graphs, adding more relations to the graph may not always improve learning, rather, it may introduce noise or shortcut structure.

This project investigates relational inductive bias of message passing - whether relational message passing improves prediction on an event-centric crime knowledge graph and how sensitive the performance is to the choice of graph relations. The main task is to predict the coarse victim-offender relationship category, which shows if there is a connection between the offender and the victim (for example, parent, boyfriend/girlfriend, and others). This leads to our research questions:

- **RQ1:** Does an R-GCN model improve over non-graph baselines for predicting victim-offender relationship categories?
- **RQ2:** Which relation types contribute useful signal, and can removing relations improve performance?

The implementation includes both non-graph baselines (majority, logistic regression baseline) and graph-based models (R-GCN). Ablations are made where specific relationship types are omitted. Furthermore, we train and evaluate ULTRA[2] as an alternative model with link prediction on knowledge graphs. We hypothesize that relational message passing improves over tabular baselines because the KG has typed neighborhood structure. However, we also hypothesize that adding all available relations may not be optimal, because the KG contains many local single entities and additional relations may weaken useful messages.

2 Related Work

Knowledge graphs are mostly used to represent data that has many different types of entities and relations. Instead of using tabular representation (each example as one row in a table), a knowledge graph models entities as nodes and their connections as typed edges. This is useful for relational data because models can use the structure between entities when making predictions. Event-centric knowledge graphs are relevant for this project because crime incidents can be treated as event nodes connected to victims, offenders, offenses, and relationship records [4].

In a standard graph neural network (GNN), a node updates its representation by aggregating information from its neighbors using message passing [3]. Since the crime KG has different relation types, a relational GCN is more appropriate than a standard GCN. R-GCNs use relation-specific transformations, so messages from different edge types (for example, "hasRelationship" and "hasIncident") can be treated differently [5]. This makes R-GCNs a suitable choice for testing whether relational message passing helps on the crime KG.

Instead of classifying relationship nodes directly, one can make use of knowledge graph completion methods to predict missing links in the graph. Models such as ULTRA approach this as a ranking problem over possible target entities [2]. In this project, ULTRA is only used as an additional comparison which can be found in Appendix C.

3 Methodology

3.1 Data and task

We use a crime knowledge graph constructed from a subset of FBI NIBRS incident data[1] for Washington state in the United States of America. The subset spans only December 2024 and is filtered for violent crimes - Murder and Nonnegligent Manslaughter, Robbery, and Aggravated Assault. The data contains around 27,000 incidents for 2024, while the subset used has around 1430 incidents with 2068 victims and 1791 offenders. As mentioned, the graph is event-centric, meaning that entities are event-specific, possibly due to anonymization of the data. This makes the dataset useful for exploring whether graph neural networks are beneficial when many entities have limited recurrence, but it is also a strong limitation of the dataset. The KG was constructed by mapping the relational data into a defined ontology (that was created for another project) and contains the following main node types: Incident, Victim, Offender, Offense, VictimOffenderRelationship, VictimType, OffenseType, and RelationshipType. The main relation types include victimInIncident, offenderInIncident, hasVictim, hasOffender, hasVictimType, involvedInOffense, hasOffenseType, hasIncident, relatedToOffender, and hasRelationshipType. A simple diagram can be seen in Figure 1.

The prediction task is to infer the victim-offender relationship category for each VictimOffenderRelationship node. In the KG, this label is represented by the relation "hasRelationshipType". The original relationship labels are highly imbalanced, with "Relationship Unknown" (Number 18) dominating the labels. Since this class mostly reflects missing or uncertain information, we remove it from the main prediction task and group the remaining labels into six categories: acquaintance_or_known, family, intimate_partner_or_spouse, stranger, victim_was_offender, and other. Full class distribution can be seen in Table 2. For R-GCN training, all hasRelationshipType edges are removed from the message-passing graph to prevent target leakage. The task

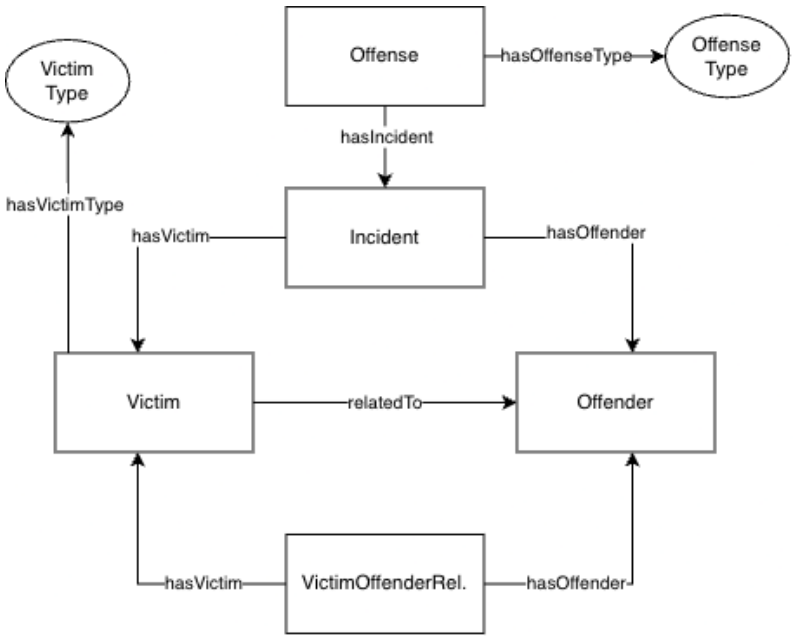


Figure 1: Simple schema of the ontology

is transductive node classification, meaning that all nodes and non-target edges are available during training, while validation and test relationship labels are masked. This means that we test whether relational structure is useful within the constructed KG, not whether the model can generalize to completely unseen incidents. Since the graph contains typed directed relations, inverse edges are added for the message passing. Data properties such as age, sex, and incident hour are used only for the tabular baseline.

We use the evaluation metrics accuracy, macro-F1, and weighted-F1, where macro-F1 is the primary metric because of class imbalance.

3.2 Experimental setup

We compared non-graph baselines, graph neural network models, and relation ablations that will be discussed in Section 4.2. All of the models were implemented on the Jupyter server of the Technical University of Vienna, equipped with an NVIDIA V100 GPU with 32GB VRAM.

The first baseline implemented is a majority classifier, which simply predicts the most frequent class in the training set. This checks how much of the task can be solved by class imbalance alone. The second baseline is a tabular logistic regression model using local features extracted from the KG, such as victim and offender sex, victim and offender age, victim type, incident hour, the number of offenses involving the victim, and the first offense type. Categorical features are one-hot encoded, numerical missing values are median-imputed, and class balancing is used during training.

The main graph model is a relational graph convolutional network (R-GCN). Nodes are initialized with trainable embeddings, and messages are propagated along typed relations using relation-specific transformations. The final embedding of each VictimOffenderRelationship node is passed to a linear classifier. The model is trained with weighted cross-entropy to handle class

imbalance. We use a hidden dimension of 64, two R-GCN layers, dropout of 0.2, Adam optimization with learning rate 0.01 and weight decay 10^{-4}, and train for 200 epochs. Results are reported over three random seeds.

4 Results

After removing the "Relationship Unknown" class and filtering very rare labels, the final task contains 1,354 victim-offender relationship records across six classes. The split contains 946 training, 204 validation, and 204 test samples. As mentioned, due to the imbalance of the data, we mainly use macro-F1 as the primary evaluation metric.

4.1 Baseline comparison - RQ1

Table 1 compares the majority baseline, the tabular logistic regression baseline, and the R-GCN models. The majority classifier predicts the most frequent training class, "stranger", and reaches an accuracy of 0.309 but only 0.079 macro-F1. This confirms that using only accuracy is misleading in this task.

The logistic regression baseline improves greatly over the majority classifier, reaching 0.373 accuracy and 0.313 macro-F1. This shows that local non-graph features such as age, sex, victim type, incident hour, and offense type already contain useful signal for the relationship prediction task. However, the full-graph R-GCN improves further, reaching 0.525 accuracy and 0.371 macro-F1 over the three random seeds. This suggests that relational message passing adds useful information beyond the manually extracted tabular features.

The best result is obtained by the pruned R-GCN graph - the demographics/core-only variant. This model removes "relatedToOffender", "involvedInOffense", and "hasOffenseType", while retaining the core victim, offender, incident, and victim-type structure. It achieves 0.544 accuracy and 0.400 macro-F1, outperforming the full graph and showing lower variance across seeds.

Table 1: Main test results. The R-GCN results are reported as mean ± standard deviation over three random seeds.

Model	Accuracy	Macro-F1	Weighted-F1
Majority baseline	0.309	0.079	–
Tabular log. regression	0.373	0.313	0.390
R-GCN full graph	0.525 ± 0.034	0.371 ± 0.049	0.498 ± 0.072
R-GCN demographics	$\mathbf{0.544 \pm 0.008}$	$\mathbf{0.400 \pm 0.004}$	$\mathbf{0.544 \pm 0.008}$

4.2 Relation ablations - RQ2

To address RQ2, we trained R-GCN variants after removing selected relation types. The full results can be found in Table 3. The full graph contains 34,098 directed message-passing edges

after adding inverse relations. As we can see, removing relations does not always reduce performance. The best-performing model is the demographics/core-only graph, which uses only six base relation types and 23,276 directed edges.

Removing only "relatedToOffender" does not hurt performance compared with the full graph. The full graph reaches 0.371 macro-F1, while the model without "relatedToOffender" reaches 0.368 macro-F1. This suggests that the R-GCN is not simply relying on this direct victim-offender shortcut edge. Similarly, removing offense context ("involvedInOffense" and "hasOffenseType") gives almost the same score as the full graph. This indicates that offense-related context does not clearly improve prediction in this setting.

However, the removal of "hasVictimType" obtains the lowest macro-F1, 0.347, and the lowest weighted-F1, 0.467. This suggests that victim type is one of the most useful relations for this prediction task. In contrast, removing "offenderInIncident" gives a slightly higher mean macro-F1 than the full graph, but with high variance across seeds. Therefore, this should not be seen as strong evidence that offender-incident edges are harmful.

5 Discussion

Overall, the results support the first research question: relational message passing improves over both the majority baseline and the tabular logistic regression baseline for our crime dataset. However, the improvement is only moderate - the best R-GCN reaches 0.400 macro-F1, so the task remains difficult, especially for minority classes such as "victim was offender".

The results also show that performance is sensitive to graph relation selection. In the results, we have shown that the full KG is not the best graph for message passing. A smaller graph focused on core victim, offender, incident, and victim-type structure performs better and is more stable across seeds. This suggests that adding all available KG relations can introduce noise rather than useful signal, especially in an event-centric KG. One possible explanation for the weaker full-graph result is that additional relations increase the size of the message-passing neighborhood without necessarily adding useful information for this task. Offense-related edges may introduce noise to the more informative local structure around victims, offenders, incidents, and victim types. Therefore, the result should not be interpreted as showing that richer KGs are, in general, worse, but that relation usefulness is task-dependent.

The main finding is that "hasVictimType" appears important, while offense-related relations do not clearly help. Therefore, the results suggest that graph construction is not just a preprocessing step but a central modeling decision for event-centric knowledge graphs.

However, the event-centric nature of the KG also limits the benefit of message passing. Many victim, offender, and incident nodes appear only in local event contexts, so the graph contains less recurring entity structure than typical benchmark KGs. This may explain why R-GCN improves over tabular features, but only moderately. Future work should explore additional months or regions to determine whether the observed importance of victim type and the weak contribution of offense context remain stable or change. Furthermore, the NIBRS dataset provides a lot of information that can be useful. Further work could extend the KG with the data tables that were omitted, such as weapons, properties, arrests, location types, to test whether these additional relations provide a useful signal or introduce further noise.

References

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[4] Simon Gottschalk and Elena Demidova. *EventKG: A Multilingual Event-Centric Temporal Knowledge Graph*. 2018. arXiv: 1804.04526 [cs.CL]. URL: https://arxiv.org/abs/1804.04526.

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A Code availability

The code is provided in an anonymized Github repository using Anonymous4OpenScience. The code can be found at: https://anonymous.4open.science/r/miniproject-46B3/

B AI statement

The author made use of ChatGPT and Grammarly to enhance the work's academic writing style and structure. The AI tool Codex was used to debug and optimize code if needed. The author carefully reviewed every suggestion made by the tools and edited the content as needed.

C ULTRA experiment

For the ULTRA experiment, the victim-offender relationship task was changed to a knowledge graph completion problem. The model was trained to rank candidate tails for triples of the form ((VictimOffenderRelationship, hasRelationshipType, ?)). The training graph contains the training triples and their inverse edges, while validation and test hasRelationshipType triples are used for evaluation. A ULTRA implementation from Practical 4 of the course was adapted and trained from scratch on the crime KG. Negative samples were restricted to valid relationship-type entities, and for evaluation the model ranked the correct relationship type among 24 candidate relationship labels. The performance was evaluated using MR, MRR, and Hits@k. The model used four layers with hidden dimension 32, Adam optimization with learning rate (10^{-3}), weight decay (10^{-5}), batch size 32, and was trained for 30 epochs over three random seeds. Model selection was based on validation MRR.

The ULTRA-style model slightly improves ranking quality over the frequency baseline, increasing MRR from 0.472 to 0.483 and Hits@3 from 0.510 to 0.574. However, Hits@1 remains unchanged at 0.309, meaning that the model does not improve top-ranked prediction over the class-frequency prior. Therefore, these results show only limited benefit from KG completion for this crime KG.

D Class distribution

Table 2: Class distribution after removing unknown and very rare relationship labels.

Class	Total examples	Test examples
Stranger	422	63
Acquaintance or known	394	59
Intimate partner or spouse	290	44
Family	191	29
Other	32	5
Victim was offender	25	4
Total	1354	204

E R-GCN relation signals

Table 3: R-GCN relation ablation results. Edges count shows directed edges after adding inverse relations. Results are mean ± standard deviation over three seeds.

Variant	Edges	Accuracy	Macro-F1
Demographics/core only	23,276	$\mathbf{0.544 \pm 0.008}$	$\mathbf{0.400 \pm 0.004}$
No `offenderInIncident`	30,516	0.531 ± 0.046	0.379 ± 0.056
Full graph	34,098	0.525 ± 0.034	0.371 ± 0.049
No `relatedToOffender`	29,836	0.528 ± 0.043	0.368 ± 0.055
No offense context	27,538	0.518 ± 0.043	0.367 ± 0.047
No `victimInIncident`	29,962	0.520 ± 0.056	0.366 ± 0.073
No `involvedInOffense`	30,428	0.508 ± 0.049	0.365 ± 0.047
No `hasOffenseType`	31,208	0.493 ± 0.061	0.364 ± 0.044
No `hasVictimType`	29,962	0.498 ± 0.028	0.347 ± 0.035